Wenfang Sun | Résumé

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Education

University of Amsterdam **Netherlands**
Ph.D. Student, Multimodal Foundation Models *2025.10–Present*
Advisor: Prof. Cees Snoek

University of Science and Technology of China **China**
Master, Computer Science *2021.09-2024.08*

Anhui University **China**
Bachelor, Communication Engineering *2017.09-2021.07*

Research Experience

Westlake University **China**
Research Assistant, Advisor: Prof. Yefeng Zheng *2025.01–2025.09*

Selected Research

Few-shot Learning with Fewer Tasks **ICML 2023**
Supervision of Dr. Xiantong Zhen and Prof. Cees Snoek.
⋆ Proposed MetaModulation, an innovative approach for few-shot learning, tackles the constraint of limited meta-training tasks by employing neural networks to modulate batch normalization parameters during training.

⋆ Proposed a variational extension of MetaModulation, introducing uncertainty-aware meta-learning through treating modulation parameters as latent variables. ⋆ Introduced learning variational feature hierarchies within the framework of variational MetaModulation, enabling modulation of features at all network layers.
⋆ Proposed an effective ensembling strategy that merges diverse segmentation maps from generated subclass descriptors, ensuring a comprehensive representation of unique characteristics in test images.

IPO: Interpretable Prompt Optimization for Vision-Language Models **NeurIPS 2024**
Supervision of Yingjun Du and Prof. Cees Snoek.
⋆ Proposed a simple yet interpretable prompt optimizer that utilizes Large Language Models (LLMs) to generate textual prompts dynamically.

⋆ Introduced a novel prompt mechanism that guides LLMs in creating effective prompts while storing past prompts with their performance metrics, providing rich in-context information.
⋆ Incorporated a large multimodal model (LMM) to generate image descriptions, enhancing the interaction between textual and visual modalities.

The Curse of Depth in Large Language Models **NeurIPS 2025**
Supervision of Prof. Yefeng Zheng and Dr. Shiwei Liu.
⋆ Revisited and formalized the *Curse of Depth* phenomenon in modern LLMs, revealing that deeper layers contribute little to training despite architectural capacity.

⋆ Provided theoretical and empirical analysis demonstrating that Pre-Layer Normalization (Pre-LN)

leads to exponential growth of output variance with depth, resulting in near-identity gradients in deep Transformer layers.
⋆ Proposed LayerNorm Scaling (LNS), a simple yet effective method that scales normalized outputs inversely with depth, mitigating variance explosion and enhancing deep-layer training.

RegionReasoner: Grounded Multi-Round Visual Reasoning **ICLR 2026**
Supervision of Yingjun Du and Prof. Cees Snoek.
⋆ Introduced RegionDial-Bench, a new multi-round visual reasoning benchmark covering both detection and segmentation tasks, enabling systematic evaluation of iterative and grounded visual reasoning beyond single-step paradigms.

⋆ Proposed RegionReasoner, a reinforcement learning framework that enforces grounded multi-round reasoning by explicitly linking each reasoning step to reference bounding boxes, while maintaining semantic coherence via a global-local consistency reward.
⋆ Demonstrated that RegionReasoner-7B significantly improves multi-round reasoning accuracy, spatial grounding precision, and global-local semantic alignment on detection and segmentation tasks, establishing a strong baseline for grounded iterative visual reasoning.

Publications

⋆ [1] **Wenfang Sun***,Yingjun Du*, Xiantong Zhen, Fan Wang, Ling Wang and Cees Snoek. MetaModulation: Learning Variational Feature Hierarchies for Few-Shot Learning with Fewer Tasks (**ICML 2023**).

⋆ [2] **Wenfang Sun***, Yingjun Du*, Gaowen Liu, Ramana Rao Kompella, and Cees Snoek. Training-Free Semantic Segmentation via LLM-Supervision (**CVPR 2024 Workshop**).

⋆ [3] Yingjun Du*, **Wenfang Sun***, and Cees Snoek. IPO: Interpretable Prompt Optimization for Vision-Language Models(**NeurIPS 2024**).

⋆ [4] **Wenfang Sun***, Xinyuan Song*, Pengxiang Li*, Lu Yin, Yefeng Zheng, Shiwei Liu. The Curse of Depth in Large Language Models (**NeurIPS 2025**).

⋆ [5] **Wenfang Sun**, Yingjun Du, Gaowen Liu, Yefeng Zheng and Cees Snoek. QUOTA: Quantifying Objects with Text-to-Image Models for Any Domain(**WACV 2026**).

⋆ [6] **Wenfang Sun**, Hao Chen*, Yingjun Du, Yefeng Zheng and Cees Snoek. RegionReasoner: Region-Grounded Multi-Round Visual Reasoning(**ICLR 2026**).

Awards

2024: Outstanding Graduate Award of University of Science and Technology of China USTC
2023: National Scholarship USTC

Reviewer

TMLR, CVPR 2026, ICLR 2026, WACV 2026, AAAI 2026, NeurIPS 2025, ICML 2025, ICLR 2025, IJCAI 2025, ACM MM 2024, ACM MM 2025.